Financial Derivatives
ES32079

Lecture 4

Forward and Futures Prices

Aliya Kenjegalieva

Structure of this lecture

Today we cover the following topics:

1. Introduction
2. No-Arbitrage Forward Pricing
3. Valuing Existing Forward Contracts
4. Applications
5. Review

Reference: Hull chapter 5.

Investment and Consumption Assets

Investment assets

Assets held by significant numbers of investors primarily for investment, such as shares, bonds, gold, and silver.

Consumption assets

Assets held mainly for use or consumption, such as oil, copper, wheat, and cocoa.

Why the distinction matters

For an investment asset, cash-and-carry arbitrage often enforces a precise relationship between spot and forward prices. For a consumption asset, inventory may provide operational benefits, so the relationship may involve a convenience yield and need not be enforced by the same arbitrage strategy.

Why Trade Forward or Futures Contracts?

Forward and futures contracts allow market participants to transfer price risk across time.

- A hedger can lock in a future purchase or sale price.
- A speculator can take a view on a future price without initially purchasing the underlying asset.
- An arbitrageur can exploit inconsistent spot and derivative prices.
- Observable futures prices can contribute to price discovery.

Important distinction

A futures price reflects today's market-clearing price for future delivery. It is not necessarily an unbiased forecast of the future spot price.

Why Is Short Selling Important?

Short selling allows investors to benefit from, or hedge against, falling prices.

- Shares are borrowed and sold in the market.
- The short seller later repurchases the shares and returns them.
- The short seller must compensate the lender for dividends and other distributions.
- Borrowing fees, collateral requirements, and recall risk may apply.
- A short position can suffer an unlimited loss if the asset price rises without bound.

Pricing relevance

Some no-arbitrage arguments require the ability to short the underlying asset. If short selling is restricted, only one side of the pricing relationship may be enforceable.

Short-Selling Example

An investor trades 500 shares initially priced at \$120. The share pays a \$1 dividend and is worth \$100 when the position is closed.

Cash flow	Buy shares	Short shares
Initial share transaction	−\$60,000	+\$60,000
Dividend	+\$500	−\$500
Closing share transaction	+\$50,000	−\$50,000
Net profit	−\$9,500	+\$9,500

Interpretation

Ignoring financing and stock-borrowing costs, the long and short profits are equal in magnitude and opposite in sign.

Forwards and Futures: Similar Obligation, Different Trading

Both contracts create an obligation to transact at a specified future date.

Feature	**Forward**	**Futures**
Venue	Usually OTC	Exchange traded
Terms	Customised	Standardised
Counterparty risk	Managed bilaterally	Managed through a clearinghouse
Settlement	Usually at maturity	Marked to market daily
Margin	Contract dependent	Required by clearing rules

Terminology

The **delivery price** K is fixed in an existing contract. The **forward price** is the delivery price that would make a new contract worth zero today.

Daily Settlement of Futures

For N long contracts, each covering Q units, the daily gain is

$$\Delta V_t = NQ(F_t - F_{t-1}).$$

- Initial margin is deposited when the position is opened.
- Daily gains and losses enter the margin account through variation margin.
- If the balance falls below the maintenance margin, a margin call may require the account to be replenished.

Example

Two long contracts cover 100 units each. If the futures price falls from \$52 to \$50.50,

$$\Delta V = 2 \times 100 \times (50.50 - 52) = -\$300.$$

Pricing Assumptions

The basic model assumes that:

- there are no transaction costs or taxes;
- assets can be traded in divisible quantities;
- market participants can borrow and lend at the relevant risk-free rate;
- short selling is possible when required;
- contractual cash flows and carrying costs are known; and
- arbitrage opportunities are exploited.

Role of assumptions

The formulas describe theoretical no-arbitrage benchmarks. Funding spreads, bid-ask spreads, collateral, taxes, and short-sale constraints can create a range around the benchmark in real markets.

Notation

- S_t: spot price of the underlying at time t
- $F_{t,T}$: forward price at time t for delivery at T
- K: delivery price written into an existing contract
- $T - t$: time remaining to delivery
- r: continuously compounded risk-free rate for the remaining term
- I: present value of known cash income before delivery
- q: continuously compounded income yield
- u: proportional storage-cost rate
- y: convenience yield

At inception

For a fairly priced new forward contract, $K = F_{0,T}$ and its value is zero. After inception, K is fixed but the market forward price changes.

Forward Price with No Income

Consider an investment asset that pays no income before delivery.

Cash-and-carry strategy

At time 0:

1. borrow S_0;
2. buy one unit of the asset; and
3. short a forward contract for delivery at T.

At T, deliver the asset for $F_{0,T}$ and repay $S_0 e^{rT}$.

No arbitrage requires

$$\boxed{F_{0,T} = S_0 e^{rT}}.$$

This is a cost-of-carry result, not a forecast of S_T.

If the Market Forward Price Is Too High

Suppose $S_0 = \$40$, $r = 5\%$, $T = 0.25$, and the quoted forward price is \$43. The no-arbitrage price is

$$F^*_{0,T} = 40e^{0.05\times 0.25} = \$40.50.$$

Cash-and-carry arbitrage

1. Borrow \$40 and buy the share.
2. Short the forward at \$43.
3. At maturity, deliver the share and receive \$43.
4. Repay \$40.50.

The locked-in profit at T is

$$\$43.00 - \$40.50 = \boxed{\$2.50}.$$

If the Market Forward Price Is Too Low

Suppose instead that the quoted forward price is \$39.

Reverse cash-and-carry arbitrage

1. Short one share and receive \$40.
2. Invest \$40 for three months.
3. Enter a long forward contract at \$39.
4. At maturity, the investment is worth \$40.50.
5. Pay \$39 for the share and return it to the lender.

The locked-in profit at T is

$$\$40.50 - \$39.00 = \boxed{\$1.50},$$

before stock-borrowing costs.

What If Short Selling Is Restricted?

Cash-and-carry can still rule out a forward price that is too high:

$$F_{0,T} \leq S_0 e^{rT}$$

under the basic no-income assumptions.
Ruling out a forward price that is too low requires the reverse cash-and-carry strategy, which normally involves shorting the asset.

Consequences

If investors cannot short the asset and do not already own it, the lower-bound arbitrage may be unavailable. The equality

$$F_{0,T} = S_0 e^{rT}$$

is therefore not guaranteed by that argument alone.

For an investment asset held by many investors, existing owners may still sell the asset and buy it forward, helping enforce the relationship.

Forward Price with Known Cash Income

Suppose the asset pays known cash income before delivery with present value I. Owning the asset provides both the terminal asset and the interim income.
The prepaid cost of the terminal asset is therefore $S_0 - I$, giving

$$\boxed{F_{0,T} = (S_0 - I)e^{rT}}.$$

Interpretation

Known income reduces the forward price because the forward holder does not receive income paid before delivery.

If the income occurs at several dates,

$$I = \sum_j I_j e^{-r_j t_j},$$

using the appropriate zero rate for each payment date.

Known-Income Example

A bond is priced at \$900. It will pay a \$40 coupon after four months. A nine-month forward is considered. The four-month zero rate is 3%, and the nine-month zero rate is 4%, both continuously compounded.
The coupon's present value is

$$I = 40e^{-0.03\cdot(4/12)} = \$39.60.$$

Therefore,

$$\begin{aligned} F_{0,T} &= (900 - 39.60)e^{0.04\cdot(9/12)} \\ &= \boxed{\$886.60}. \end{aligned}$$

Check

Why is the forward price below the spot price even though interest rates are positive?

Arbitrage Around the Known-Income Price

The fair forward price is \$886.60.

If quoted at \$910	If quoted at \$870
Borrow to buy the bond and short the forward. Use the coupon to service the corresponding borrowing. At maturity, deliver the bond for \$910 and repay \$886.60. Profit: \$23.40.	Short the bond and invest the proceeds after setting aside the present value of the coupon compensation. Go long the forward at \$870. At maturity, the investment provides \$886.60. Profit: \$16.60.

Practical qualification

The reverse strategy requires the bond to be available for borrowing and the short seller to compensate the lender for the coupon.

Forward Price with a Known Yield

If the asset provides a continuously compounded income yield q, the no-arbitrage forward price is

$$\boxed{F_{0,T} = S_0 e^{(r-q)T}}.$$

Example

Let $S_0 = 25$, $r = 10\%$, $q = 3.96\%$, and $T = 0.5$. Then

$$F_{0,T} = 25e^{(0.10-0.0396)\times 0.5} = \boxed{\$25.77}.$$

Cash income versus yield

Use $(S_0 - I)e^{rT}$ for known cash amounts and $S_0 e^{(r-q)T}$ for income proportional to the asset's value.

Forward Price Is Not Forward Value

At inception, a fairly priced forward has

$$K = F_{0,T} \qquad \text{and} \qquad f_0 = 0.$$

At a later time t:

- the contractual delivery price K remains fixed;
- the market forward price becomes $F_{t,T}$; and
- the contract value f_t is generally no longer zero.

For a long forward,

$$\boxed{f_t = (F_{t,T} - K)e^{-r(T-t)}}.$$

For a short forward, the value is $-f_t$.

Deriving the Value of a Long Forward

Construct a portfolio at time t:

1. Keep the existing long forward with delivery price K.
2. Enter a new short forward with delivery price $F_{t,T}$. A new fairly priced forward has zero value.

At delivery, the combined payoff is certain:

$$(S_T - K) + (F_{t,T} - S_T) = F_{t,T} - K.$$

Discounting this certain payment gives

$$f_t = (F_{t,T} - K)e^{-r(T-t)}.$$

For a no-income asset, substituting $F_{t,T} = S_t e^{r(T-t)}$ gives the equivalent expression

$$\boxed{f_t = S_t - Ke^{-r(T-t)}}.$$

Valuing an Existing Forward: Example

A long forward on a non-dividend-paying share has delivery price $K = \$24$. Six months remain. The current share price is \$25 and the continuously compounded risk-free rate is 10%.
The current forward price is

$$F_{t,T} = 25e^{0.10\times 0.5} = \$26.28.$$

The value of the existing long forward is

$$\begin{aligned} f_t &= (26.28 - 24)e^{-0.10\times 0.5} \\ &= \boxed{\$2.17}. \end{aligned}$$

Equivalently,

$$f_t = 25 - 24e^{-0.10\times 0.5} = \$2.17.$$

Forward and Futures Prices

Forward and futures prices are equal when interest rates are deterministic and the contracts otherwise have the same underlying and delivery date. When rates are uncertain, daily settlement can create a difference:

- If futures gains tend to occur when interest rates are high, those gains can be reinvested at favourable rates, tending to make the futures price exceed the forward price.
- If futures gains tend to occur when rates are low, the opposite tendency arises.

Key mechanism

The difference comes from the timing and reinvestment of daily variation-margin cash flows, not from a different terminal payoff formula.

Stock-Index Futures

A stock index can be treated as an investment asset paying a dividend yield q. Under the standard assumptions,

$$\boxed{F_{0,T} = S_0 e^{(r-q)T}}.$$

Example

For a three-month contract, let

$$S_0 = 1{,}300, \quad r = 5\%, \quad q = 1\%, \quad T = 0.25.$$

Then

$$F_{0,T} = 1{,}300 e^{(0.05-0.01)\times 0.25} = \boxed{1{,}313.07}.$$

The dividend yield used should correspond to dividends expected during the life of the contract.

Currency Forwards: Covered Interest Parity

Let S_0 be the domestic-currency price of one unit of foreign currency. Holding foreign currency allows the investor to earn the foreign interest rate r_f, while domestic funds earn r_d.
No arbitrage therefore requires

$$\boxed{F_{0,T} = S_0 e^{(r_d - r_f)T}}.$$

Quotation matters

The signs depend on how the exchange rate is quoted. Here both S_0 and $F_{0,T}$ are domestic currency per unit of foreign currency.

If $r_d < r_f$, the foreign currency trades at a forward discount under this quotation convention.

Currency-Forward Example

Suppose

$$S_0 = 0.9800 \text{ USD per AUD}, \quad r_{USD} = 1\%, \quad r_{AUD} = 3\%, \quad T = 2.$$

The no-arbitrage forward rate is

$$\begin{aligned} F_{0,T} &= 0.9800 e^{(0.01-0.03)\times 2} \\ &= \boxed{0.9416 \text{ USD per AUD}}. \end{aligned}$$

Prediction

If the market forward rate is only 0.9300 USD per AUD, which currency should the arbitrageur borrow?

Currency Arbitrage When the Forward Is Too Low

If the two-year forward rate is 0.9300 USD per AUD:

1. Borrow 1,000 AUD and convert it to 980 USD.
2. Invest 980 USD at 1% for two years, producing $980e^{0.01\times 2} = \$999.80$.
3. The AUD loan grows to $1{,}000e^{0.03\times 2} = 1{,}061.84$ AUD.
4. Buy 1,061.84 AUD forward for $1{,}061.84 \times 0.9300 = \$987.51$.
5. Use the AUD to repay the loan.

The riskless profit at maturity is

$$\$999.80 - \$987.51 = \boxed{\$12.29}.$$

Commodities: Storage Costs and Convenience Yield

Physical commodities may involve:

- financing cost r;
- storage and insurance costs u; and
- convenience yield y, the non-cash benefit of holding inventory.

If proportional carry components are used,

$$\boxed{F_{0,T} = S_0 e^{(r+u-y)T}}.$$

If storage costs are known cash amounts with present value U, and there is no convenience yield,

$$\boxed{F_{0,T} = (S_0 + U)e^{rT}}.$$

Consumption assets

Convenience yield can make holding physical inventory more valuable than holding only a forward contract. Reverse cash-and-carry arbitrage may therefore be unavailable.

The Cost of Carry

The net proportional cost of carrying an asset to delivery is

$$c = r + u - q - y,$$

where terms that do not apply are set to zero. Then

$$F_{0,T} = S_0 e^{cT}.$$

Underlying	Net cost of carry
Non-dividend-paying investment asset	r
Stock index	$r - q$
Foreign currency	$r_d - r_f$
Commodity	$r + u - y$

Interpret carefully

Cost of carry explains the spot-forward relationship; it does not imply that the future spot price must equal today's forward price.

Commodity Forward: Known Storage Cost

Consider a one-year contract on an investment asset with:

$$S_0 = \$450, \qquad r = 7\%,$$

and a \$2 storage payment due at the end of the year.
The present value of the storage cost is

$$U = 2e^{-0.07} = \$1.865.$$

Therefore,

$$\begin{aligned} F_{0,T} &= (450 + 1.865)e^{0.07} \\ &= \boxed{\$484.63}. \end{aligned}$$

If the quoted forward price is above \$484.63, cash-and-carry arbitrage is possible under the model assumptions.

Contango and Backwardation

Contango The forward price exceeds the spot price.

Backwardation The forward price is below the spot price.

From

$$F_{0,T} = S_0 e^{(r+u-q-y)T},$$

contango is associated with positive net carry, while a sufficiently large income or convenience yield can produce backwardation.

Do not infer too much

Contango does not by itself mean that the market expects the spot price to rise, and backwardation does not by itself mean that the market expects it to fall. Carry and risk premia also matter.

Choose the Correct Pricing Formula

Match each situation to its pricing formula.

1. A non-dividend-paying share
2. A bond paying a known coupon before delivery
3. A stock index with dividend yield q
4. A foreign currency
5. A commodity with storage cost and convenience yield

1	$S_0 e^{rT}$
2	$(S_0 - I)e^{rT}$
3	$S_0 e^{(r-q)T}$
4	$S_0 e^{(r_d - r_f)T}$
5	$S_0 e^{(r+u-y)T}$

Lecture Summary

- Forward prices follow from spot prices and the net cost of carry.
- Cash-and-carry addresses an overpriced forward; reverse cash-and-carry addresses an underpriced forward when short selling is possible.
- Known cash income and proportional yields require different formulas.
- The value of an existing forward is not the same as its current forward price.
- Daily settlement can create a difference between futures and forward prices when interest rates are uncertain.
- Currency pricing follows covered interest parity.
- Commodity pricing may include storage costs and convenience yield.

Unifying principle

Replicate future delivery using spot-market borrowing, lending, ownership, and income. If the derivative price differs from the replication cost, investigate the arbitrage and the assumptions it requires.